[FULLY FICTITIOUS DOCUMENT · FOR FAIRPANEL LIVE DEMO] All entities, metrics, and profiles are created for multimodal OCR and fair assessment demonstration.

CANDIDATE IDENTIFIER: FP-C02 · TECHNICAL ARCHITECTURE PORTFOLIO

Doyun Kim — FinTech Systems Architecture Portfolio

- **Architecture 1:** Legacy 6-Hour to 18-Minute Kafka Streaming Settlement Engine Migration
- **Architecture 2:** 3-Week Shadow-Traffic Benchmark Resolution & Multi-Engineer Consensus
- **Architecture 3:** Distributed Redisson Lock & Idempotency Key Pipeline Design
- **Performance Metric:** Settlement processing time slashed from 360 min to 18 min (95% improvement)
- **Reliability Standard:** 100% duplicate charge prevention under concurrent retry surges

System 1. Event-Driven Real-Time Settlement Engine

System 01 **Replacing Monolithic Batch Processing with Event-Driven Streaming Architecture**

• Background: Legacy monolithic settlement cron jobs took over 6 hours to reconcile nightly merchant ledgers, creating cascading operational delays for customer payout calculations.
• Architecture: Designed a decoupled, distributed stream processing pipeline utilizing Apache Kafka partition keys mapped to merchant routing groups, backed by PostgreSQL append-only ledgers.
• Performance: Cut total settlement execution duration from 360 minutes to 18 minutes while supporting linear throughput scaling up to 10,000 events/sec.
• Integrity: Implemented idempotent message processing with SHA-256 transaction hash keys, eliminating duplicate journal entry vulnerabilities.

Scope Boundary: Banking network gateway protocol agreements were handled by external financial network providers.

System 2. 3-Week Shadow-Traffic Benchmark & Technical Alignment

System 02 **Resolving Conflicting Migration Approaches Among Senior Engineering Leads**

• Conflict: Two staff engineers favored an in-memory Redis cluster for settlement caching, while another advocated for direct database write-behind batches due to persistence concerns.
• Resolution Methodology: Refused subjective argumentation; engineered an objective 3-week shadow-traffic benchmarking harness mirroring 100% of live transaction traffic to both candidate architectures.
• Outcome: Benchmark data proved that the event-streamed write-behind queue guaranteed zero data loss during simulated cluster partition tests, uniting the senior engineering team on empirical evidence.

Scope Boundary: Demonstrates collaborative, data-driven technical leadership without organizational friction.